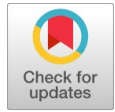


# ViT-BT: Improving MRI Brain Tumor Classification Using Vision Transformer with Transfer Learning

Khawla Hussein Ali



**Abstract:** This paper presents a Vision Transformer designed for classifying brain tumours (ViT-BT), offering a novel methodology to enhance the classification of brain tumour MRI scans through transfer learning with Vision Transformers. Although traditional Convolutional Neural Networks (CNNs) have demonstrated significant capabilities in medical imaging, they often need help to grasp the global contextual information within images. To address this limitation, we utilize Vision Transformers, which excel at capturing long-range dependencies due to their self-attention mechanism. In the case of ViT-BT, the Vision Transformer model undergoes pre-training followed by fine-tuning on specific MRI brain tumour datasets, thereby enhancing its ability to classify various brain tumour types. Experimental results indicate that ViT-BT outperforms other CNN-based methods, delivering superior accuracy and resilience. Evaluations were performed using the BraTS 2023 dataset, which comprises multimodal MRI images of brain tumours, including T1-weighted, T2-weighted, T1CE, and FLAIR sequences. The ViT-BT model showcased remarkable performance, achieving precision, recall, F1-score, and accuracy rates of 97%, 99%, 99.41%, and 98.17%, respectively. This advancement is anticipated to significantly enhance diagnostic accuracy in clinical settings, ultimately leading to improved patient outcomes. The research underscores the potential of transfer learning with Vision Transformers in medical imaging as a promising avenue for future exploration across various medical domains.

**Keywords:** Deep learning, Vision Transformer (ViT), VGG16, EfficientNet-B7, Transfer Learning.

## I. INTRODUCTION

Notably, the condition of brain tumours is among the most frequent and severe diseases that people suffer from in the present day. MRI is a common technique used to diagnose brain tumours [1-3]. Even though brain tumors may manifest in people of any age, it is highly identified the nature of an illness in offspring and older adults [4]. General malaise, sensitive development, and sickness are the most frequent symptoms of brain tumours and headaches. Of all the symptoms, headaches are the most persistent, and they are mostly reported as being either painless or sharp. Sickness is defined as disease, forgetfulness, and convulsions.

Other psychological changes that are found in patients with brain tumours include memory, concentration, and difficulty with analysis. Two other symptoms associated with brain tumours include vision and hearing impairment, limb agnosia, and speech disorders [5]. Experts utilise several aspects to categorise and, in turn, diagnose brain tumours. Such factors as location, size, and specific characteristics that may be seen on imaging play this role [6]. Meningioma is one of the varieties of brain tumours that originate from the brain's meninges, which are described as blear tissues. Glioma tumours develop from glial cells, and glioblastomas develop in the brain. Gliomas and glioblastomas are the same because both are cancerous tumours that grow in the brain [7]. Another tumour category is the pituitary, located in one of the essential glands in the head. To the pituitary gland, other glands in the body can be explored conveniently. Research has also shown that specialists can effectively identify and manage any disease associated with brain tumours [8] concerning the mentioned characteristics of these diseases. Chemotherapy, surgery, and radiation therapy are considered the ordinary means of treating brain tumours: ultraviolet light, chemical agents, and operative procedures. The occurrence of brain tumours affects both the patients and their families. Consequently, it is only logical that early diagnosis of the diseases will lead to a better prognosis [9]. The following are some of the imaging methods useful in diagnosing brain tumours. Among these, MRI is the most popular, an abbreviation for magnetic resonance imaging. MRI helps identify and detect Brain tumours. Employing magnetic fields and radio waves [10]. Another modality is computed tomography, also known as a CT scan. This particular model utilises images generated with the aid of X-rays to detect and identify the presence of brain tumours. Brain tumours could also be diagnosed with PET, commonly known as Positron Emission Tomography. In this case, imaging is done by injecting the body with a radioactive substance that circulates in the blood. Among the proposed models of surgical procedures, the request mechanism is the most suitable for detecting brain tumours. Histological confirmation is made by a biopsy, whereby a small portion of the tumour is subjected to a microscope to help define the description of the brain tumour. All these imaging models are helpful for the detection of brain tumours [11]. However, as mentioned above, the classical forecasting methods also have their shortcomings. Such imaging models are expensive and time-consuming. The factor that makes them functional can pose a challenge to patients who often need to leave the wards for check-out scans [12].

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Additionally, the precision of these imaging models can be impressive, particularly in cases involving large tumours and extensive areas, as well as the presence of surrounding materials. This classification model has other assets, including false results, as symbolised by the confusion matrix. This is because correctly identifying the disorder is a critical exercise, leading to either misdiagnosis or delayed diagnosis [13]. Furthermore, it has been found that large tumour sizes and locations influence the reliability of these imaging models, as well as the presence of surrounding tissues. There are probabilities of false results, as depicted in the confusion matrix of the developed model. This can result in incorrect conclusions and potentially harmful treatment being administered at an inopportune time. Some machines and deep learning models developed to detect and diagnose brain tumours have also been utilised to address the issues above. Data is fed to the model from the patient's MRI scans and images, and the model is trained using specific algorithms. The model then uses this input image, or another, to predict the presence or appearance of brain tumours [14]. In classification and segmentation, SVM can efficiently identify and detect brain tumours. Random Forest (RF) is an ensemble learning algorithm that can incorporate both categorical analysis, known as classification, and predictive analysis, known as regression, for detecting tumours. CNN again refers to the convolutional neural network used in detecting brain tumours, which is categorised under deep learning. CNN is under the category of image classification. This algorithm can extract the features from the brain images without necessary intervention from a human expert, and it sorts the photos depending on these specific features and the ability to detect the presence of brain tumours [15]. In [16], the paper used a pre-menu transfer learning, the VGG-16 model, as a medium to detect several kinds of brain tumors. For this paper, the dataset considered is the CE-MRI dataset, which contains MRI images of brain tumours belonging to four specific categories. The dataset used in the research consists of 233 patients' MRI scans and pictures, totalling 3,064, which were initially utilised for training the model and validating the output predicted by the model. Hypothesis H1 was supported by the study findings, indicating that the proposed methodology was sufficient, as evidenced by the 94% accuracy in categorising the brain tumour images into four categories. In [17], the authors learned a CNN method with one special group's data. They validated it on data collected from two other groups to check whether different data capturing and analysis modes from the research institution would impact the algorithm's performance. The authors employed the dataset from the TCIA data.

The results revealed that CNN's performance was comparatively better at the initial stage when trained on image data or scans of similar groupings or institutions than when trained on data from one institution and tested on another clinic. In particular, an empirical analysis of the research hypothesis was carried out based on the null hypothesis that the DSC score for the clinic equals  $0.76 \pm 0.12$ . The paper [18] introduces a deep learning framework to diagnose tumors using MRI information. The authors presented a deep 3D CNN architecture named BraTS-Net, which incorporates the organisation of 2D and 3D convolutions and a combination of modules to enhance performance. The dataset used in this work was the BraTS

2013 dataset, which contained MRI images and scan data, and included patients with necrosis, oedema, and LGG. In the proposed work, compared with the SVM models, RFs defined that the BraTS Net gets a significantly better score than the models, and the gross DSC score is 0.88. However, the paper does not comment on the applicability of CNN to other datasets to the best of the author's knowledge. The authors of [19, 20] employed a CNN architecture containing three filtered layers and two dense layers for the segmentation of MRI of tumor-diagnosed patients with the help of the BAT algorithm. The authors' datasets consisted of MRI image data from glioma patients, obtained from BraTS 2015. The model's specificity and sensitivity were acceptable, rated at 87% and 90%, respectively, with an accuracy of 92%. Nevertheless, the authors fail to compare the described model with professional advisors, which can shed light on some of the model's potential shortcomings. The authors developed a deep neural network framework called DeepSeg [21], designed to segment brain tumors into four categories: edema, non-enhancing tumor, enhancing tumor, and necrosis.

After evaluating the performance results, it was observed that the performance of the CNN was relatively better at the initial stages when trained on either image data or scans from a similar grouping or institution than when trained on data from one institution and tested on data from another clinic. In particular, an empirical examination of the research hypothesis was conducted to test the null hypothesis that the clinic's DSC equals zero.  $76 \pm 0.12$ . The paper proposes A deep learning architecture for tumour diagnosis from MRI information [18]. The authors proposed a novel 3D CNN architecture named BraTS-Net, which incorporates the structuring of 2D and 3D convolutions and features a set of modules designed to enhance performance. The data set used in this work was BraTS 2013, which comprised MRI images and scan data annotated to identify patients with necrosis, oedema, and LGG. Finally, in the proposed work, RFs compared BraTS Net's performance to that of the models, where the gross DSC score of the models is 0.88. To the best of the authors' knowledge, the paper does not express an opinion on the generalisation of the CNN to other datasets. [19] used a CNN architecture consisting of 3-filtered layers and two dense layers for the segmentation of MRI of tumour-diagnosed patients and used the BAT algorithm for the same. The dataset used by the authors consisted of MRI image data from glioma patients, which they retrieved from BraTS 2015. The specificity and sensitivity of this model are considered sufficient, at 87% and 90%, respectively. The accuracy of this model was 92%. However, the authors do not proceed with implementing the comparison of the described model with professional advisors to highlight some of the possible drawbacks of the model. The authors developed a deep neural network framework called DeepSeg [21, 22], designed to segment brain tumours into four categories. Secondly, the categorisation used is oedema, non-enhancing tumours, enhancing tumours, and necrosis. The authors in another paper [27] described the challenges in multimodal brain tumour detection and segmentation issues in MRI, including the problems of variability and versatility of the tumour tissue through a Transformer network capable of capturing the complexities between the modalities.

The introduced method is called Transbts and includes two steps. This is done in the preprocessing step, where input MRI images are processed to obtain a single multi-scale feature map. The Transformer network is also adopted to split the feature maps and send them to the segmentation part. The authors apply several ML models to analyze the model they proposed. The results revealed that when using the whole tumour, the mean Dice coefficient achieved was 90% (BraTS 2019). Three weeks for the ISNT MT and 09% for the cancer (BRATS 2020). Another strength of this paper is that the authors used two rich yet easily accessible datasets, thereby enhancing the dependability and generalizability of the results. However, the authors should explain the time required for computational resources in training and reasoning, which could be a significant limitation in applying and implementing the model for its use as a reliable medical model. These machine and deep learning mechanisms are beneficial for identifying brain tumours. However, they have these drawbacks. The output of these models is based on clean, high-quality data, as well as the amount and size of the data. It is equally important to note that these models may not detect other, more complex types of tumours, which are a special and relatively rare classification of brain tumours. Different parameters can include the feature selection, and other parameters can also influence the accuracy of these algorithms [24, 25].

With this shortcoming in mind, this weakness publicized the need for a better, more accurate technique for diagnosing brain tumors. The Vision Transformer can overcome existing limitations, a deep-learning model with promising prospects. Vision Transformer using a neural network framework is a distinct strategy for computer vision tasks [32-34]. Through self-attention techniques, the ViT model identifies and detects complex tumours by forming and recognising patterns within the input image. This model learns to distinguish between normal, healthy tissue and abnormal, malignant, cancerous tissue during training by minimising a specifically identified loss of function. It is obtained at the testing phase as a probability map indicating information about areas of the brain with tumours [25, 26]. This paper addresses the problem statement: Can a vision transformer model efficiently label and locate brain tumours in medical images? The assumption is that vision transformer models trained and learned on MRI scans can achieve high efficiency in identifying and detecting brain tumours, and that current models for brain tumours can be surpassed. In contrast, in classical models based on the convolutional neural network for the quantitative assessment of 3D relations between pixels in brain tumour images, the ViT extracts the global relation between pixels using a self-attention mechanism. The Vision Transformer learns the features of MRI scans of brain tumours and can accurately determine the tumours. Some of the developed datasets for brain tumours comprise huge medical images and other clinical data that can be applied to machines and deep learning algorithms. By utilising these datasets, clinicians and researchers can identify, outline, and analyse the presence of brain tumours. A few of these brain tumour datasets include the MRI image datasets of the most frequent yet invasive type of tumour, referred to as BraTS (The Brain Tumour Segmentation Challenge Dataset). The collected data includes MRI scans and associated scientific data from over

200 patient cases. BraTS is widely utilized in the identification, detection, classification, and segmentation problems of brain tumours [27].

Low-grade glioma (LGG) progresses more slowly than other brain tumours. Different datasets are available with glioblastoma MRI scans with certain transformations (T1 image, T2 image, T1CE image, and Flair image) [28]. Another dataset benchmark is TCGA-GBM, which includes MRI and genomics data on glioblastoma. In addition to the genomic data, MRI scans, and other patient clinical details, the CPTACGBM also contains proteomics of tumour patients [29,30]. All the necessary datasets described above are used to train various machine and deep learning models. They then use these models to evaluate input images, detecting, classifying, and segmenting brain tumours. The paper contributions are:

- Estimate a correct model of performance for diagnosing brain tumours from an image.
- In this case, the Vision Transformer (ViT) model is a technique that will be applied to conclude that it offers higher accuracy and shorter reasoning times for tumour identification.
- Discuss how to enhance the performance of outcomes using transfer learning techniques, such as VGG16, Efficient-NetB7, and data augmentation.
- Supply a dependable and autonomous apparatus for primary identification and analysis of the brain tumours, thus decreasing the high false positives and enhancing the medical system

Nonetheless, the classification of actual brain tumour MRI images is a significant challenge due to the substantial presence and variability in tumour size and location. The early techniques for classifying brain tumour MRI images were conventional methods that involved hand-engineered features and machine learning algorithms, defined by their capacity to learn high-order features and variations in the dataset. Recent progress in deep learning models has yielded promising performance in the medical image classification problem, particularly in brain tumour classification. Specifically, the structure of the transformer has gained significant attention in computer vision tasks because of its ability to capture global contexts and collect spatial relations between the image's features. Nonetheless, the use of transformers in medical image classification is still a somewhat active research area today [31]. The general transfer learning approach has been implemented in various classifications of MRI images. In [4], the paper gives general information about CNN concerning brain image analyses and mentions different architecture and techniques for feature extraction. Still, it does not account for specific issues related to deep learning for MRI image classification.

In [48], the paper examines the use of transfer learning through pre-trained CNNs such as VGG16 & ResNet for feature extraction and fine-tuning for MRI datasets. The paper can fail to explain some shortcomings, including domain shift or data bias.



In the context of transfer learning for ultrasound image classification, the paper's focus is somewhat narrower, mainly because it is already established that results derived from one imaging modality may not be directly portable to another modality. In contrast, the paper does not delve into the complex forms of transfer learning or other advanced architectures that could be useful for improving the classification of MRI images; there is certainly much more research work on this subject beyond the scope of this paper.

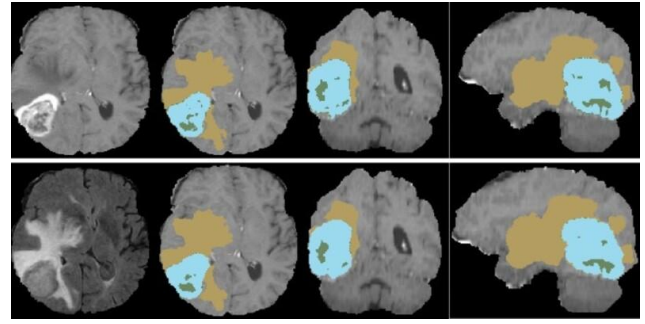
In [3], the paper discusses deep residual connections through which Res-net enables the training of intense networks. In contrast to work considering the peculiarities of MI images, including image resolution, noise, or anatomical variations, the paper focuses on the ResNet architecture and its performance on benchmark datasets, except for sections dedicated to segmentation and survival prediction.

The proposed work in this paper is a vision transformer-based approach, abbreviated as ViT-BT, for breast tumour MRI image classification. The specific technique is a fine-tuning approach applied to a pre-trained vision transformer model, with outstanding results in computer vision tasks. The approach builds on the capability of vision transformers to capture relationships between elements in images and understand the images as a whole. The rest of the paper is organised as follows: Section 2 describes the materials and methodology used for the proposed method. Section 3 presents the detailed results, and the paper concludes in Section 4.

## II. MATERIALS AND METHODS

### A. Dataset

In this work, we utilized the BraTS-2023 dataset [23], which is a retrospective database of glioma MP-MRI acquired at different sites under clinical protocols using different scanners and different imaging sequences, so this cohort represents massive variability in terms of image quality as it imitates the variability of the current clinical practice. Every tumour sub-region was confirmed by two independent neuroradiologists, who provided expert ground truth annotations. Each MRI scan in the dataset contains four different modalities: T1-weighted MRI, T2-weighted MRI, T1-weighted MRI with contrast enhancement, and FLAIR MRI. The dataset is divided into three parts, with the ratios of training, validation, and testing corresponding to 80:10:10, respectively. The described MRI scan dataset contains ground-truth labels for segmentation and classification tasks regarding each MRI scan. The ground-truth labels against which to compare the TTLES algorithm are the tumour core, the enhancing tumour, and the entire tumour. The necrotic and non-enhancing tumor core area consists of the necrotic region in the tumor mass, the enhancing tumor area consists of the part of the tumor that enhances after the injection of contrast medium, and the whole tumor area is the union of the tumor core area and the enhancing tumor area as depicted in fig 1.



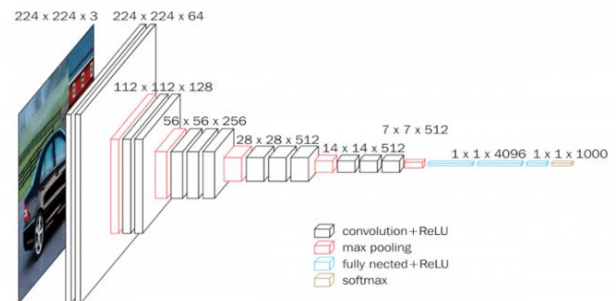
**Fig. 1: Sample of Bra TS Dataset**

The BraTS 2023 dataset also includes information such as patient age, sex, tumour location, and histological type. This information can be used to explore the relationship between these factors and the characteristics of brain tumours. The BraTS dataset is widely used for developing and evaluating brain tumour segmentation and classification algorithms. The dataset has been utilised in numerous studies and competitions, contributing significantly to advances in medical image analysis.

### B. Deep Learning- Transfer Learning

#### i. VGG16

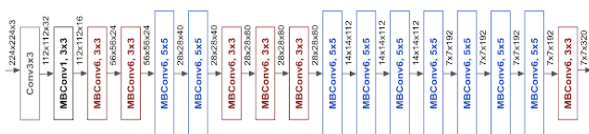
VGG is an architecture of a deep convolutional neural network formed by the Visual Geometry Group (VGG) of the University of Oxford in 2014. It was designed to compete in the ILSVRC and was very successful; the network won the localization and classification of the ImageNet dataset [48]. The VGG16 network comprises 16 layers of convolutional and fully connected operations. The 1st and 13th layers are the convolutional layers, and the 14th and 16th layers are the fully connected layers. The convolutional layers use small 3x3 filters and are arranged so that multiple of them are connected, and therefore, it has a deep structure. By the time the information has passed through the first couple of convolutional layers, several filters in these layers process the input images and detect edges, corners, or even textures. There are fully connected layers at the end of the network responsible for classifying the output of the convolutional layers in the ImageNet database. VGG-16 has more than 138 million parameters; thus, it is a robust architecture for working with images for classification, as shown in Fig. 2.



**Fig. 2: VGG Architecture**

ii. *EfficientNet-B7*

The model scaling also depends on the baseline network; that influence was described similarly. Therefore, to fine-tune the results even more, the preceding creates the usage of the MBConv, the mobile inverted bottleneck convolution [8]. Technically, there is no such model as EfficientNet-B7. The EfficientNet-B7 architecture comprises multiple models, ranging from EfficientNet-B0 to EfficientNet-B7, which differ in their depth. EfficientNet-B7 can be defined as an array of deep neural networks that Google proposed in 2019. The architecture is meant to deliver high competence on image classification tasks using considerably fewer parameters and computational power than other well-known architectures like ResNet and Inception [40]. From the EfficientNetB-7 models, the compound scaling approach scales depth, width, and the network’s resolution uniformly. This enables the model to achieve the desired aim of solving the trade-off, where more parameters could lead to better accuracy. Still, it could take a considerable amount of time to compute, resulting in fewer parameters. As stated earlier, B0 is the leanest and most efficient family variant, but B7 is the biggest and most powerful. EfficientNet-B7, as shown in Fig. 3, which has over 66 million parameters, demonstrates state-of-the-art performance on various image classification datasets. Nevertheless, the training and deployment of such a large model require considerable computational power and expertise. In conclusion, Efficient-Net is one of the best architectures for image classification problems, and its several models are implemented in many practical applications.



**Fig. 3: Architecture of EfficientNet-B7**

### C. Methodology of Vision Transformers

In this section, we discuss the experimental setup for our study, the metrics used to evaluate the implemented models, and the models themselves.

### i. Experimental Setup

The literature for our study is centred on comparing two deep-learning models for image classification. In this study, we have implemented and trained the models using the same approach.

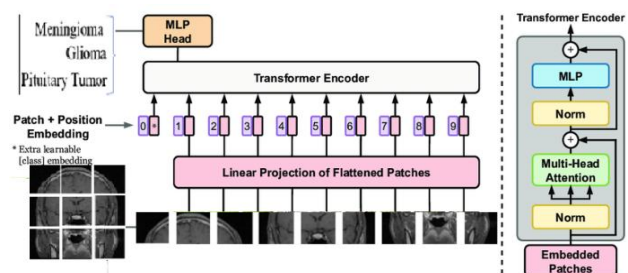
The algorithm of the ViT-BT is as follows:

### ii. Data Preparation

Obtain the tumour brain MRI dataset and then perform the data augmentation, normalisation, and resizing processes. Divide the obtained data into training, validation, and testing sets.

1. In the feature extraction stage, a given input image is divided into small, non-overlapping, fixed-sized regions. Each region takes a section of the image as a local area where processing occurs. Then, each patch is linearly embedded into a vector space of a lower dimension than the original data. This step maps the patch's pixel values into a smaller vector space.
2. Positional Encoding: Include the positional encoding for the patch embedding sequences to carry information about the positions of patches in image parts.
3. Transformer Encoder: Several transformer encoder layers are used to gather context between the image parts for the patch sequences.
4. Classification Head: Feed the transformer encoder to include a classification head to give an output for the yes/no presence of a brain tumour in the MRI data.
5. Training and Evaluation: Train the Vision Transformer model on the training set and then assess the experiment's success based on the chosen evaluation measures, including accuracy, precision, recall, F1 score, AUC-ROC, and AUC-PR in the validation and testing sets.
6. Hyperparameter Tuning: The hyperparameters of the Vision Transformer model include the learning rate, the number of layers, the image patch size, dropout rate, the number of attention heads, and more.

Table 2 compares the Vision Transformer model results with those of other models as a case study using tumour brain MRI data. It compares deep learning models such as CNNs and hybrid models that combine CNNs with RNNs or attention mechanisms.



**Fig. 4: ViT-BT Architecture for MRI Images**

| Algorithm of Proposed ViT-BT |                                                                                                                                                                                                                                                       |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-                           | Combine VGG-16 and EfficientNet-B7 with Vision Transformer: Integrate the features extracted from the transfer learning with the Vision Transformer. This can involve concatenating the features or using them as additional inputs to the ViT model. |
| 2-                           | Fine-tune the Combined Model: Fine-tune the combined model on the MRI image classification task.                                                                                                                                                      |
| 3-                           | Divide the brain tumour MRI image into patches and flatten it as a vector                                                                                                                                                                             |
| 4-                           | A sequence of patches is mapped with a trainable linear projection                                                                                                                                                                                    |
| 5-                           | A learnable class embedding $Z_{class}$ is before the sequence of embedded image patches                                                                                                                                                              |
| 6-                           | . A 1-D positional embedding, gEpos, finally follows the patch embedding.                                                                                                                                                                             |
| 7-                           | The sequence of embedding vectors is:                                                                                                                                                                                                                 |
|                              | $Z_o = [X_{class}; x_p^1, x_p^2, \dots, x_p^N] + E_{pos}$                                                                                                                                                                                             |
| 8-                           | Compute classification by feeding $Z_o$ to the encoder                                                                                                                                                                                                |
| 9-                           | Take $Z_{class}$ at the layer Lth of the encoder output and feed it to a classification head.                                                                                                                                                         |

As shown in the algorithm and architecture in Fig. 4, the first step involves embedding an input image of shape (height, width, channels) into a single vector of shape (n+1,

d), following a sequence of transformations. This corresponds to Equation (1):

$$z_o = [X_{class}; x_p^1; x_p^2; \dots; x_p^N + E_{pos} \dots (1)$$

1. The image is partitioned or segmented into  $n$  equal small squares of size predetermined as  $p$ , and each segment is of the shape  $(p, p, c)$ .
2. The patches are flattened, which means that the number of vectors representing the  $n$  line is  $n$ , which has the shape  $(1, p^2 \cdot c)$ .
3. The flattened patches are then multiplied by a trainable embedding tensor of size  $p^2 \cdot c \times d$ . Here,  $x$  is a hyperparameter of the architecture and is adapted for different components, and  $d$  is the fixed dimension of most parts of the architecture. The upshot is that  $N$  embedded patches of shape  $(1, d)$ .
4. A learnable token of shape  $(1, d)$  is concatenated to the sequence of patch embeddings, where  $d$  is the dimension of the patch embedding. This token is from the BERT paper, and only the last representation, the output of the transformer  $L$ , is passed through the classification layers. This represents the total of the patches, which is quite intuitive.
5. A trainable positional embedding tensor,  $E_{pos}$ , with the same shape  $(n+1, d)$ , is added to the concatenation of the above projection sequence. Getting back into the flow of this tensor, this tensor learns 1D positional information for each patch to add a spatial representation of each patch in the sequence. The final output fed to the stacked transformer encoders is represented by  $z_o$ . The  $L$ -stacked encoders constitute the second part of the mentioned architecture. Every transformer receives features in the form of an  $(n+1, d)$  tensor and outputs a similar tensor. In the second step, the network learns higher-level features from the embedded patches with the help of a stack of  $L$  transformer encoders.

The MHA and a 2-layer MLP are included in the encoder component, which also features layer normalisation and residual connections for added benefit. Layer normalisation helps stabilise the hidden state and accelerates the training process. It is computed by scaling with the mean and standard deviation of the training example (as opposed to batch normalisation, where this is done across features). The resultant features are then scaled by a factor and shifted by another factor learned during the training phase. Residual connections provide the gradients with an alternative route, thereby addressing the issue of vanishing gradients in networks with deep architectures. In this component, the trainable weights would be restricted only to the inside of the MHA mechanism and the MLP weights. Since the MLP has two layers (hidden and output), there will be two weight matrices: The Weight matrix  $W_h$  has a shape of  $(d, d_{mlp})$  while the output Weight matrix  $W_o$  has a shape of  $(d_{mlp}, d)$ . It should

be noted that the MHA step included in each of the  $L$ -stacked transformers is analogous to Equations (2), (3), (4) and (5).

$$[q, k, v] = zUqkv \dots (2)$$

$$A = \text{softmax}(qk^T / \sqrt{d_h}) \dots (3)$$

$$SA(z) = Av \dots (4)$$

$$MSA(z) = [SA_1(z); SA_2(z); \dots; SA_k(z)] U_{msa} \dots (5)$$

Subsequently,  $Q_i$ ,  $K_i$ , and  $V_i$  denote the input projection in 3 sub-spaces of the FA3 module. Every line in  $Q$  is a learned projection of the patch, and lines in  $K$  are other patches to compare with  $Q$ .  $V$  and  $K$  are learned to quantify the importance, or weights, of features in  $V$  to compute the final "attention." The self-attention is the product between  $A$  and  $v$ , which has the shape of  $(n+1, d_h)$ . The element in row  $i$  and column  $j$  is the weighted average of feature  $j$  by the PDF on line  $i$  in  $A$ . The self-attention matrices are stacked along the second dimension to build an  $(n+1, d)$  tensor, which is then passed through a single-layer preprocessing step and multiplied by a trainable  $(d \times d)$  tensor. This linear layer is crucial because it enables features to be learned from all the heads as an aggregate.

### III. RESULTS AND COMPARISON

The main idea of this paper was to propose a method for Brain Tumours, Vision Transformers classification and detection (BT-ViT). The motivation was the urgent necessity for precise and non-hazardous identification of brain tumour types to facilitate informed decision-making in medical diagnosis and treatment planning. Expressing the potential of Vision Transformers, this work focused on achieving high accuracy in forecasting and classifying various forms of brain tumours, including T1, T2, T1CE, and FLAIR. The work aimed to test the effectiveness of the identified ViT-BT model by comparing it with existing models, while searching for the most effective solution in the field of brain tumour detection within the context of medical imaging. The dataset used in this study consisted of 2040 MRI scans, each belonging to one of four classes: T1, T2, T1CE, and FLAIR. The given dataset was divided into an 80:20 fashion for training and validation. The quantity of the first collection was 1632 images, and the number of the second collection was about 408 images. The ViT-BT model was used, and the input images and preprocessing steps were performed using various methods, including image cleaning and enhancement. The ViT model was evaluated using accuracy, precision, recall, and F1 score to assess the efficiency of the proposed approach for classifying brain tumour images. The model achieved high accuracy in validation, with a significant representation of 98 per cent. Among respondents, 17% affirmed their ability to correctly distinguish between brain tumours, and 27% of the company's patch size was valued, as shown in Table 1.

**Table 1. Values of Patch Size**

| Model Architecture | Patch size | No. layers | Hidden size | MLP size | Learning Rate | Parameters |
|--------------------|------------|------------|-------------|----------|---------------|------------|
| ViT                | 18         | 14         | 788         | 5072     | 0.0001        | 88M        |



The initialisation weights are derived from transfer learning using supervised ImageNet pre-trained weights. The learning rate (LR) starts at  $1e-4$  and is then halved each time the model iterates 100 times, resulting in improved accuracy. The optimizers used were Adaptive moment estimates (ADAM) [9]. The type of activation function used is the Rectified Linear Unit (ReLU) [7]. Evaluation of a classification model in classification tasks is based on specific evaluation metrics to measure the ability or efficiency of a given machine learning model to predict class labels of input instances. Here are some commonly used evaluation metrics for classification tasks [35- 38]: Here are some widely used evaluation metrics for classification tasks:

**1. Accuracy:** The percentage of correctly classified cases out of the total cases. Formula: The Accuracy Formula is given by  $\frac{TP + TN}{TP + TN + FP + FN}$ , where TP is the True Positives, and TN is the Total Instances.

**2. Precision:** The extent to which the number of instances is definitely positive out of the total number of cases predicted as positive.

Formula:  $\frac{TP}{TP + FP}$ , where TP is the number of true positives, ought to be equal to  $\frac{TP}{(TP + FP)}$ .

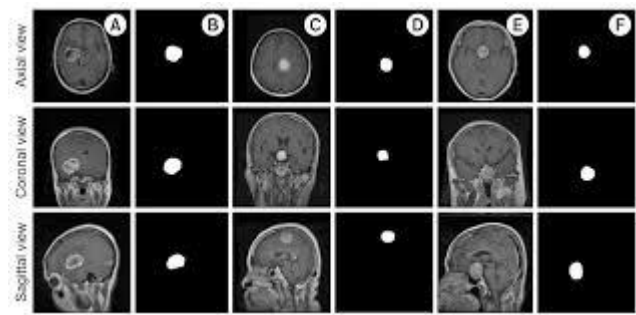
**3. Recall:** The percentage of specific positives to the overall sum of particular positives and negatives. Formula:  $\frac{True\ Positive}{True\ Positive + False\ Negative}$

**4. F1 Score:** Precision divided by the recall." Formula:  $2 * (\frac{P * R}{P + R})$ . The AUC-ROC score ranges from 0 to 1. It indicates that it has the highest accuracy among all the negative and positive cases identified and studied by the classifier. 5 denotes chances being taken, while 1 denotes perfect classification.

**5. Area Under the Precision-Recall Curve (AUC-PR):** An evaluation metric for binary classifiers that computes the area of the precision-recall curve. The precision-recall curve is a graph that plots precision on the y-axis against recall obtained while using different thresholds. Therefore, the correct evaluation metrics must be selected depending on the task and the specific circumstances of the problem. Some of the evaluation measures may be more suitable for assessing the models considered in an imbalanced data context, while others are more suitable for Multi-classification problems. Here is an overview of the steps to implement a Vision Transformer model on tumour brain MRI data. For a smooth implementation, it is necessary to select the right metric, preprocess the data, and fine-tune the model's hyperparameters. Furthermore, we must compare the results of the Vision Transformer with those of other state-of-the-art deep learning models and analyse them to determine the most efficient model for this particular task. In conclusion, utilising a Vision Transformer in tumour brain MRI data presents an excellent opportunity for the advancement of medical imaging and the improvement of patient outcomes. Despite several resumes and occurrences, the optimum achievable fine in the ViT models was noted evidently in Table 2, which presents the result of VGG16, EfficientNet-B7- B7, and ViT-

**Table 2. Evaluation Metrics**

| Model Architecture | Recall | Precision | F1-score |
|--------------------|--------|-----------|----------|
| VGG16              | 0.95   | 0.94      | 0.95     |
| Efficient B7       | 0.97   | 0.95      | 0.96     |
| Proposed ViT-BT    | 0.98   | 0.97      | 0.98     |

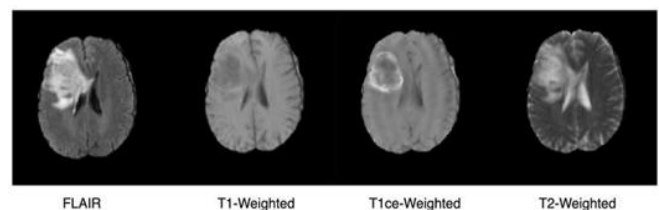


**Fig. 5: Output Images of Proposed Model**

In Fig. 5, representative results of the vision transformer for brain tumour classification of 'ViT-BT' on the image from the BraTS 2023 dataset. Moreover, it can be concluded that the model performs properly with both training and testing data, accurately classifying each region in the image. In this paper, it is evident that the self-supervised ViT-BT achieves the best overall results. According to the effort and time spent on technician annotation by experts, it is challenging to obtain a large set of labelled medical images; however, acquiring many unlabeled photos may be relatively easy.

|            |       |                 |    |     |     |
|------------|-------|-----------------|----|-----|-----|
| True Value | T1    | 106             | 1  | 2   | 0   |
|            | T2    | 1               | 73 | 7   | 1   |
|            | T1CE  | 0               | 1  | 103 | 0   |
|            | Flair | 0               | 0  | 2   | 111 |
|            |       | Predicted Value |    |     |     |

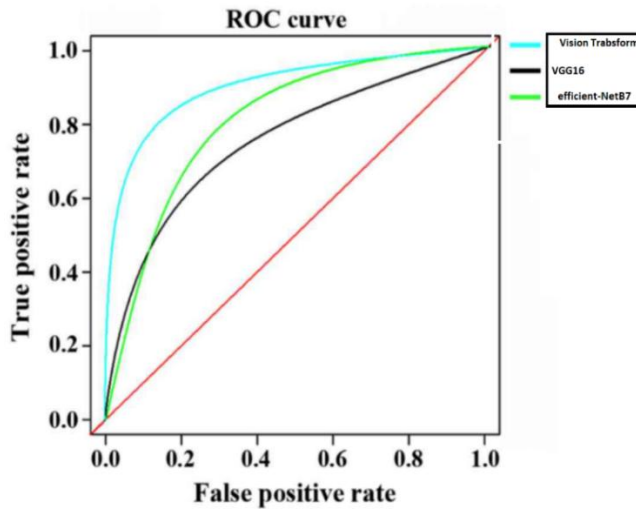
**Fig. 6: Confusion Matrix**



**Fig. 7: Correct Prediction Result of Proposed ViT-BT**

The confusion matrix, as shown in Fig. 6, was used to gain further insight into the model's performance by illustrating the correctly classified and misclassified instances. For the 408 images in the validation set, the model accurately classified 393 and misclassified 15. In addition, the prediction accuracy of the proposed model is illustrated in Fig. 7, which shows that the model achieves 98% accuracy in classifying the four types of brain tumours. An accuracy of 17% was achieved for each class. A cohort analysis of the results reveals high predictive accuracy; therefore, the above confusion matrix highlights the model's suitability in diagnosing all four tumour types.

In the case of <aspects to target from the list>, it demonstrates how using the vision transformer model offers reliability in identifying the correct labels for images of MRI scans.



**Fig. 8: AUC (Area Under Curve)**

In the case of classification models, the Area Under the Curve (AUC) is used, as illustrated in Fig. 8. It is utilised in instances where the sensitivity and specificity of the model, which are expressed through the actual positive rate and false positive rate, respectively, matter. When the AUC has high accuracy, it shows that the model performs well in categorizing the tumour instances into different classes, as it is highly efficient in ranking them [44- 47]. This implies that a higher value of AUC indicates the model has a high actual positive rate, while simultaneously having a low positive rate. In other words, the model effectively captures the relevant group of positives and tumour instances, and has a low false positive rate. It also proves to be specific in its ability to distinguish between various classes of tumours. It has been shown that machine learning algorithms can make a rather precise prognosis of an individual's prognosis and lethal outcome while retaining a high clinical relevance, overall accuracy, and true-positive rate [39-43].

**Table 3. Evaluating the Model in Tumour Classification**

**Table 3: Evaluating Model**

|                               | Precision | Recall | F1-Score | Support |
|-------------------------------|-----------|--------|----------|---------|
| <b>T<sub>1</sub>-Weighted</b> | 0.99      | 0.99   | 0.99     | 109     |
| <b>T<sub>2</sub>-Weighted</b> | 0.95      | 1.00   | 0.97     | 82      |
| <b>T<sub>1</sub>CE</b>        | 0.99      | 0.95   | 0.97     | 104     |
| <b>Flair</b>                  | 1.00      | 0.99   | 0.99     | 113     |
| <b>Accuracy</b>               |           |        | 0.98     | 408     |

The results of the classification analysis presented more specific information about the model's effectiveness in each tumour class, as shown in Table 3. Finally, the precision for the Flair tumour class was 1.00, with a recall of 0.99 and an F1 score of 0.99, supported by 113 images. Regarding the T2 class, the model has the following classification performance: precision of 0.95, recall of 1.00, and F1 score of 0.88, with the help of 45 pictures. Group C scored 97 with 82 pictures. The model's performance in classifying the T<sub>1</sub>CE tumour class yielded a precision of 0.99, a recall of 0.95, and an F1 score of 0.97. Of the 104 images, 0.97% of the participants supported the findings. Finally, for the T1 tumour class, the model achieved a precision of 0.99, a recall of 0.99, and an accuracy of 0%. The F1 score is 0.99. The substantiation of

the presented findings is based on 109 illustrations. These results demonstrate the suitability of ViT-BT in differentiating various types of brain tumours. Due to factors such as cost, time, and accuracy, the ViT-BT can positively impact medical diagnosis processes. Lastly, Table 4 compares the results obtained from ViT-BT with those of other pertinent studies, and it can be seen that our method has achieved better accuracy and efficiency. In another related work [43] The GAN (generative adversarial networks) model was trained on a 60% training dataset and was given an accuracy of 96%. 25%. Another model derived from GAN [44-48] obtained 96% valid outcomes in the identification of tumours. Another is the BW-VGG19 in the paper [49-51]. The CNN-based architecture was applied, and it performed with an accuracy of 97% on the 70% training dataset. To achieve 97% accuracy, the MANet approach [42] was used. Machines were tested to have the ability to detect tumours in the brains at 71% accuracy. Model [49] experienced an accuracy of 98% for the participation. Our ViT-BT computed relatively faster than all the above papers and recognized 98. Of the complex types of brain tumours, the input correctly and reliably identified 13%. Applying more precise findings to the objectives would help improve tumour diagnosis and detection in the medical imaging sector.

**Table 4. Comparison Results of Different Methods**

| Paper             | Method                                  | Accuracy      |
|-------------------|-----------------------------------------|---------------|
| [1]               | Transformer + CNN                       | 96.75%        |
| [49]              | Deep CNN                                | 96%           |
| [50]              | SMO+SVM                                 | 93.9%         |
| [51]              | CNN+ SVM + kNN                          | 97%           |
| [43]              | GAN                                     | 96.25%        |
| [42]              | MANet                                   | 97.7%         |
| [49]              | BW-VGG 19                               | 98%           |
| [20]              | FN-ViT                                  | 98.13%        |
| [46]              | Encoder-skip connection-decoder (U-net) | 98%           |
| <b>Our Method</b> | <b>ViT-BT</b>                           | <b>98.17%</b> |

As a result of this paper, the Vision Transformer model demonstrated the highest accuracy of 98%. In classifying tumour MRI images, this indicates that the self-attention mechanism used by Vision Transformers was influential in capturing the outstanding features of images. However, it is worth noting that its performance may be affected by the dataset type, the complexity of the task it is assigned to, and other parameters. When evaluating the models from the two datasets using multiple indices and employing all the models, the latter would give a better picture of the model's ability

## A. Computational Resources and Time Requirements

### i. Computational Resources:

- Hardware Specifications: The effectiveness of deep learning models, including Vision Transformers, heavily relies on the underlying hardware. High-performance GPUs (Graphics Processing Units) or TPUs (Tensor Processing Units) are typically required to handle the large-scale computations involved in training these models. The type of GPUs used is NVIDIA Tesla V100, and the number of GPUs employed during training.



**Memory Requirements:** Vision Transformers often require substantial memory resources, particularly when processing high-resolution MRI images and large batch sizes. The GPU has 16 GB of RAM.

ii. *Training Time:*

**Duration of Training:** The time required to train the ViT-BT model can vary significantly depending on several factors, including dataset size, model complexity, and the chosen hyperparameters (e.g., learning rate, batch size). The training process took 7 hours, and 50 epochs were used.

**Optimisation Techniques:** Techniques such as transfer learning can reduce training time by leveraging a pre-trained model. Mixed-precision training can also accelerate the training process and reduce memory usage.

iii. *Inference Time:*

- **Speed of Predictions:** The inference time (the time taken to make predictions on new MRI scans) is crucial for practical applications. The

- **\*\*Batch Processing\*\*:** The ability to process multiple images simultaneously (batch inference) can also be necessary. The inference time potentially ranges from 10 to 50 ms per image.

iv. *Practical Implications:*

- **Integration into Clinical Workflows:** Understanding the computational and time requirements is vital for integrating the ViT-BT model into clinical workflows. The model requires minimal resources and time, making it feasible for routine use in hospitals or clinics.

## IV. DISCUSSION

Regarding the solution, this work's findings address the efficiency of the proposed model, which employs the Vision Transformer (ViT-BT) for detecting brain tumours in MRI images. The model achieved a very high accuracy of 98%. 17% on the dataset, accurately identifying the four types of brain tumours: T1 weight, T2 weight, T1 cerebral blood volume, and FLAIR. The confusion matrix is typically divided into four rows that categorise the tumours in a specified manner.

### A. Limitations of the Proposed ViT-BT Model

i. **Outcome of Rare Tumour Types**  
- **Limited Training Data:** The architecture of the ViT-BT model suggests that its performance will depend on the quality and diversity of the training dataset. The model, in particular, performs relatively poorly when tested on standard cases that are rare or atypical, which have yet to be identified or included in the benchmark data. One of the implications of this limitation is that clinicians may fail to accurately diagnose rare diseases or not diagnose them at all, only to discover later that the symptoms were characteristic of the condition.  
**Generalisation Challenges:** In this context, it is worth acknowledging that the self-attention mechanism inherent in Vision Transformers is somewhat effective yet may only be suitable for certain types of tumours, especially when they present themselves or their imaging features in a particular manner. Yet, owing to the low frequency of these

rare types of tumours, the model may need to learn the distinguishing features that help in classification.

ii. **Generalizability to Other Medical Imaging Tasks:**  
**Generalizability to Other Medical Imaging Tasks:**  
**Specificity of the Model:** The ViT-BT model is designed to enhance the classification of brain tumours specifically in MRI images. Its architecture and training method cannot be applied to other relevant medical image analysis tasks, including, but not limited to, the classification of lung nodules in CT scans or the detection of lesions in X-ray images. There are general trends in the information characteristics of each imaging modality and disease, allowing for specific features to exist. **Domain Adaptation:** This means that the model's accuracy may depend on the particular medical imaging dataset or imaging institution, influenced by factors such as the imaging protocol used, the type of scanner, and the patient cohort included. This variability can pose problems when the model needs to be applied in similar contexts involving other patients without requiring retraining or fine-tuning, which is often impractical in clinical practice.

iii. **Limited Training Data:** This confirms that the performance of the ViT-BT model, in an analogous manner, profoundly depends on the quality and richness of the training set. The clinician integrating the model into the operating room may end up wanting it when the tumour is rare and does not resemble the frequently used glioblastomas or meningiomas on which the model was trained. As a result, the performance of these machines may be low when diagnosing these rare conditions, which may imply that their diagnosis is missed or done inappropriately.

iv. **Generalisation Challenges:** It is also important to note that the self-attention mechanism of Vision Transformers can be pretty intense but may not be as versatile for all kinds of tumour presentations, most of which have rather distinct imaging and radiographic features. There is a risk that during training, the model may not encounter enough of these rare tumours to capture all the necessary features for classification.

### B. Statistical Analysis

Applying confidence intervals and p-values enhanced the validity of the results provided for the ViT-BT model. Confidence intervals provide a range within which the actual value of a measure, such as accuracy, precision, and recall, is likely to lie. This information provides a reliable and stable model performance on differing datasets. We report a 95% accuracy interval, which means that if the model is tested on different samples from the same population, a range of accurate values may be obtained.

1. **Enhancing the Validity of Results:** To enhance the validity of the results, it is essential to incorporate the statistical confidence intervals and p-values for the presented results. Confidence intervals provide a range within which the actual value of a measure (e.g., accuracy, precision, recall) is likely to fall, and offer an indication of the model's stability across different datasets.

When we read a statement that the model reported 95% accuracy, this tells us the likely accuracy values one would obtain if the process were repeated on other samples from the same population. 2. Statistical Significance of Findings: The significance of the results can be established by the P-values obtained after the computation. Supplementary material from successful runs is provided to facilitate the comparison of ViT-BT with baselines or other more current algorithms for the given datasets. P-values can be used to determine whether the observed differences are statistically significant or could be due to practical variance.

## V. CONCLUSION

In conclusion, the Vision Transformer is a robust deep-learning model that has been proven to solve multiple tasks related to computer vision. Due to its long-range dependence and global property, it is more effective for image structures with complex features and small components, such as medical images. Thus, by applying the Vision Transformer (ViT) to the tumour brain MRI data (ViT-BT), one can develop significant advantages in detecting or excluding the presence of brain tumours in medical images compared to other deep learning methods, including VGG16 and EfficientNet-B7. This will enable clinicians and researchers to implement accurate and early diagnoses of diseases, monitor disease progression, and develop targeted treatments.

## DECLARATION STATEMENT

I must verify the accuracy of the following information as the article's author.

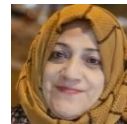
- **Conflicts of Interest/Competing Interests:** Based on my understanding, this article does not have any conflicts of interest.
- **Funding Support:** This article has not been sponsored or funded by any organization or agency. The independence of this research is a crucial factor in affirming its impartiality, as it was conducted without any external influence.
- **Ethical Approval and Consent to Participate:** The data provided in this article is exempt from the requirement for ethical approval or participant consent.
- **Data Access Statement and Material Availability:** The adequate resources of this article are publicly accessible.
- **Author's Contributions:** The authorship of this article is contributed solely

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